Video Streaming Troubleshooting

Video & Streaming - Video Streaming Troubleshooting

This guide provides solutions for common video streaming issues in GCXONE. Follow the troubleshooting steps in order, as later solutions may depend on earlier fixes.

No Video Display

Symptoms:

- Black screen or "No video available" message
- Camera shows online but no video stream
- Video player loads but shows nothing

Device Offline or Unreachable

- Verify device is powered on and connected to network
- Ping device IP address to test connectivity
- Check network cable connections
- Verify device is accessible from GCXONE server

Incorrect Credentials

- Verify username and password in GCXONE device configuration
- Test credentials by logging into device web interface
- Update credentials in GCXONE if they were changed on the device

RTSP Port Blocked

- Verify RTSP port 554 is open
- Check firewall rules allow RTSP traffic
- Verify port forwarding if accessing remotely

Stream Path Incorrect

- Verify correct RTSP path for device manufacturer
- Common paths: /stream1, /h264, /video, /live
- Check device documentation for exact path and update in GCXONE

Codec Not Supported

- Verify device codec is H.264 or H.265
- Update browser to latest version

- Try a different browser — Chrome is recommended

Poor Video Quality

Symptoms:

- Pixelated or blurry video
- Choppy playback
- Low resolution display

Low Bitrate Setting

- Access device web interface and navigate to Video/Streaming settings
- Increase bitrate — recommended 2–4 Mbps for 1080p
- Adjust frame rate if needed — 15–30 FPS is typical

Bandwidth Limitations

- Check available bandwidth
- Reduce stream quality in GCXONE to Medium or Low
- Enable adaptive quality in GCXONE settings
- Use secondary stream for grid views and limit concurrent streams

Network Latency

- Test latency by pinging the device IP
- Use Local Mode for on-site workstations
- Check network routing and switches

Frequent Buffering or Stuttering

Symptoms:

- Video pauses frequently
- Loading spinner appears often
- Playback is choppy

Insufficient Bandwidth

- Reduce stream quality and limit concurrent streams
- Close unused video windows and enable adaptive quality

Network Packet Loss

- Test packet loss by running an extended ping to device IP
- Check network cables and connections
- Verify network switch performance and check for congestion

Device Performance Issues

- Reduce number of simultaneous streams from device

- Lower stream bitrate on device
- Check device temperature and cooling
- Restart device if needed

Connection Drops or Timeouts

Symptoms:

- Video stream disconnects frequently
- "Connection lost" messages
- Stream stops after a period of time

Network Instability

- Test network stability with continuous ping
- Check network cable connections and switch/router stability
- Check for interference on WiFi connections

Timeout Settings Too Short

- Increase timeout in GCXONE streaming settings to 30–60 seconds
- Adjust device timeout settings and enable keep-alive mechanisms

Firewall Timeout

- Adjust firewall timeout settings
- Configure firewall to allow persistent connections
- Check NAT timeout settings

Device Resource Exhaustion

- Reduce concurrent connections to device
- Lower stream quality and bitrate
- Restart device and check for firmware updates

Local Mode Not Working

Symptoms:

- Local Mode enabled but still using cloud
- No P2P connection established
- High latency despite Local Mode being active

Local Mode Service Not Running

- Verify Local Mode service is running in Windows Services
- Restart Local Mode service
- Reinstall Local Mode if needed and check service logs for errors

Network Access Issues

- Verify workstation and device are on the same network
- Test connectivity by pinging device IP
- Check network routing, VLAN configuration, and firewall rules

Device Not Supporting Local Mode

- Verify device type supports Local Mode
- Note: Some devices such as Hanwha and Genesis Audio do not support Local Mode
- Use cloud streaming for unsupported devices

Auto-Discovery Failed

- Manually configure device IPs in Local Mode settings
- Check network broadcast settings
- Verify UDP ports for discovery are open and restart Local Mode service

Slow Stream Startup

Symptoms:

- Takes a long time for video to start
- Loading spinner appears for extended period
- Delay before first frame appears

High Latency Connection

- Use Local Mode for on-site access
- Check network latency by pinging device IP
- Optimize network routing

Device Slow to Respond

- Check device CPU and memory usage
- Restart device if overloaded
- Update device firmware and reduce load on device

Stream Initialization Issues

- Verify RTSP path is correct
- Test RTSP directly with VLC or a similar tool
- Update device firmware

Audio Issues

Symptoms:

- No audio with video
- Audio out of sync
- Poor audio quality

Audio Not Enabled

- Enable audio in device settings
- Enable audio in GCXONE camera configuration
- Check audio codec support

Audio Codec Not Supported

- Verify codec is G.711, AAC, or PCM
- Update browser to latest version
- Check device audio codec settings

Audio Sync Issues

- Adjust buffer settings
- Check device firmware for audio sync fixes
- Use Local Mode for better sync if available

Still Having Issues?

1. Check device logs for specific errors
2. Review GCXONE platform logs for streaming errors
3. Gather information including device type and model, firmware version, network configuration, and error messages with timestamps
4. Contact Support and submit a support ticket with the gathered information